

## SEMESTER S8

### PATTERN RECOGNITION

|                                            |                  |                    |                |
|--------------------------------------------|------------------|--------------------|----------------|
| <b>Course Code</b>                         | <b>OEEVT 831</b> | <b>CIE Marks</b>   | 40             |
| <b>Teaching Hours/Week<br/>(L: T:P: R)</b> | 3:0:0:0          | <b>ESE Marks</b>   | 60             |
| <b>Credits</b>                             | 3                | <b>Exam Hours</b>  | 2 Hrs. 30 Min. |
| <b>Prerequisites (if any)</b>              | PBECT304         | <b>Course Type</b> | Theory         |

#### Course Objectives:

1. To introduce the fundamental algorithms for pattern recognition
2. To instigate the various classification and clustering techniques

### SYLLABUS

| <b>Module No.</b> | <b>Syllabus Description</b>                                                                                                                                                                                                                                                                                                                          | <b>Contact Hours</b> |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>1</b>          | Introduction: Basics of pattern recognition system, various applications, Machine Perception, classification of pattern recognition systems<br><br>Design of Pattern recognition system, Pattern recognition Life Cycle. Statistical Pattern Recognition: Review of probability theory, Gaussian distribution, Bayes decision theory and Classifiers | <b>8</b>             |
| <b>2</b>          | Concept of feature extraction and dimensionality, Curse of dimensionality, Hidden Markov Models (HMM) - basic concepts, Gaussian mixture models.<br><br>Non-Parametric methods: Non-parametric techniques for density estimation - Parzen-window method, K-Nearest Neighbour method.                                                                 | <b>9</b>             |
| <b>3</b>          | Decision trees: Construction, splitting of nodes, choosing of attributes, overfitting, pruning<br><br>Linear Discriminant based algorithm: Perceptron, Support Vector Machines                                                                                                                                                                       | <b>9</b>             |
| <b>4</b>          | Multilayer perceptrons, Back Propagation Algorithms. Artificial Neural Networks<br><br>Classifier Ensembles: Bagging, Boosting/ AdaBoost<br><br>Unsupervised learning: Clustering - Criterion functions for clustering, Algorithms for clustering: K-means and Hierarchical methods, Cluster validation                                              | <b>10</b>            |

**Course Assessment Method**  
(CIE: 40 marks, ESE: 60 marks)

**Continuous Internal Evaluation Marks (CIE):**

| Attendance | Assignment/<br>Microproject | Internal<br>Examination-1<br>(Written) | Internal<br>Examination- 2<br>(Written ) | Total |
|------------|-----------------------------|----------------------------------------|------------------------------------------|-------|
| 5          | 15                          | 10                                     | 10                                       | 40    |

**End Semester Examination Marks (ESE)**

*In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions*

| Part A                                                                                                                                                     | Part B                                                                                                                                                                                                                                                                    | Total |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>2 Questions from each module.</li> <li>Total of 8 Questions, each carrying 3 marks</li> </ul> <p>(8x3 =24marks)</p> | <ul style="list-style-type: none"> <li>Each question carries 9 marks.</li> <li>Two questions will be given from each module, out of which 1 question should be answered.</li> <li>Each question can have a maximum of 3 sub divisions.</li> </ul> <p>(4x9 = 36 marks)</p> | 60    |

**Course Outcomes (COs)**

At the end of the course students should be able to:

| Course Outcome |                                                                                                                           | Bloom's<br>Knowledge<br>Level (KL) |
|----------------|---------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| CO1            | Design a pattern recognition system                                                                                       | K2                                 |
| CO2            | Know the major approaches in statistical and syntactic pattern recognition.                                               | K2                                 |
| CO3            | Become aware of the theoretical issues involved in pattern recognition system design such as the curse of dimensionality. | K2                                 |
| CO4            | Implement pattern recognition technique                                                                                   | K3                                 |

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

**CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)**

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| CO1 | 2   | 2   |     |     |     |     |     |     |     |      |      | 3    |
| CO2 | 2   | 2   |     |     |     |     |     |     |     |      |      | 3    |
| CO3 | 2   | 2   |     |     |     |     |     |     |     |      |      | 3    |
| CO4 | 2   | 2   |     |     |     |     |     |     |     |      |      | 3    |

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

| Text Books |                                            |                                     |                       |                               |
|------------|--------------------------------------------|-------------------------------------|-----------------------|-------------------------------|
| Sl. No     | Title of the Book                          | Name of the Author/s                | Name of the Publisher | Edition and Year              |
| 1          | Pattern Recognition and Machine Learning,  | C M Bishop                          | Springer              | 2006                          |
| 2          | Pattern Classification and scene analysis, | R O Duda, P.E. Hart and D.G. Stork, | John Wiley            | 2 <sup>nd</sup> edition, 2001 |

| Reference Books |                                                                     |                                  |                                    |                  |
|-----------------|---------------------------------------------------------------------|----------------------------------|------------------------------------|------------------|
| Sl. No          | Title of the Book                                                   | Name of the Author/s             | Name of the Publisher              | Edition and Year |
| 1               | Pattern Recognition Engineering                                     | Morton Nadler, Eric P. Smith     | John Wiley & Sons, New York        | 1993.            |
| 2               | Pattern Recognition : Statistical, Structural and Neural Approaches | Robert J. Schalkoff              | John Wiley & Sons Inc. New York    | 2007             |
| 3               | Pattern Recognition                                                 | S.Theodoridis and K. Koutroumbas | Academic Press                     | 4/e,2009         |
| 4               | Machine Learning                                                    | Tom Mitchell                     | McGraw Hill                        | 1997             |
| 5               | Pattern Recognition Principles                                      | Tou, J.T. and Gonzalez, R.C.     | Addison-Wesley Publishing Company, | 1974             |

| Video Links (NPTEL, SWAYAM...) |                                                                                                                 |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Module No.                     | Link ID                                                                                                         |
| 1                              | <a href="https://www.youtube.com/watch?v=mfePdDh9t6Q">https://www.youtube.com/watch?v=mfePdDh9t6Q</a>           |
| 2                              | <a href="https://www.youtube.com/watch?v=yG1nETGyW2E">https://www.youtube.com/watch?v=yG1nETGyW2E</a>           |
| 3                              | <a href="https://www.youtube.com/watch?v=pMHOPezBUfU">https://www.youtube.com/watch?v=pMHOPezBUfU</a>           |
| 4                              | <a href="https://youtu.be/EWmCkVfPnJ8?si=re_1ZpaXCtOohvwq">https://youtu.be/EWmCkVfPnJ8?si=re_1ZpaXCtOohvwq</a> |